

Matheus Pimentel Brackmann Caramuru

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• github.com/MatheusPBC

Professional Summary

Backend and Cloud developer with experience designing distributed systems, APIs, automation workflows, IoT telemetry, and cloud-native infrastructure. Strong background with AWS Lambda, AppSync, DynamoDB, Node.js, TypeScript, Python, MongoDB, Docker, and infrastructure as code, with focus on scalability, cost awareness, observability, and maintainability in production systems.

Experienced in high-volume data environments, including pipelines for reading, validating, transforming, and delivering field equipment measurements, as well as event-driven and serverless processing architectures. Also works with applied AI in engineering workflows and products, including RAG, MCP, LLM automation, and agents for productivity, data analysis, and new feature development.

Technical generalist with a solid foundation in backend, databases, and cloud, able to understand legacy systems, refactor complex flows, reduce bottlenecks, and turn business problems into practical solutions. Collaborative profile with experience supporting architectural decisions and technical mentoring.

Experience

SmartLy

Brazil

Backend Developer and Architect

Apr 2024 - Present

- Architected and developed a cloud-native IoT device management platform using AWS AppSync, AWS Lambda, DynamoDB, GraphQL, and serverless integrations.
- Modeled high-volume telemetry data and flows, balancing millions of messages, latency, per-operation cost, fan-out, and operational consistency.
- Designed real-time communication alternatives using GraphQL subscriptions, MQTT, WebSockets, DynamoDB Streams, Redis, interest management, and selective fan-out.
- Refactored backend structures, queries, and data contracts to reduce coupling, improve maintainability, and unlock production system evolution.
- Built and maintained APIs, Lambdas, and event-driven integrations focused on observability, rollback, security, and predictable cost.
- Investigated bottlenecks, reviewed architecture, supported technical decisions, and mentored the team through practical engineering trade-offs.

Eletrisa

Brazil

Full Stack Developer

Aug 2023 - Apr 2024

- Developed APIs, web applications, and internal systems for customers in the energy and power plant sector.
- Built complete systems with Node.js, TypeScript, React, MongoDB, Docker, and automated integrations.
- Created APIs with full CRUD flows, validation, authentication, and structured operational data.
- Developed and maintained Node-RED flows, including dashboards, integrations, automation, and real-time data processing.
- Created pipelines to read field equipment data, persist raw data, validate, transform, and periodically send data to external integrations.
- Worked with telemetry and hydrology data such as rainfall, level, flow, timestamps, extra fields, and equipment/item grouping.
- Fixed MongoDB aggregation, query performance, timestamp handling, table rendering, and dashboard issues.

Intelligence 4 Innovation

Brazil

Systems Developer / DevOps

Feb 2022 - Jun 2023

- Developed systems, flows, and applications with strong use of databases and automation.

- Supported and evolved systems connected to a large database used in a public and municipal environment.
- Developed and adapted code for continuous maintenance, support, and improvement of existing systems.
- Worked with Python, C++, GeneXus, and SQL, deepening data modeling, queries, manipulation, and production problem solving.
- Supported DevOps routines, application maintenance, and technical process improvements.

Plasnox Ind. e Com. de Plásticos Ltda

Brazil

Purchasing Assistant

Jan 2021 - Nov 2021

- Issued invoices, registered purchases, and operated enterprise management systems.
- Gained early experience with administrative processes, operational organization, and corporate routines.

Skills

- **Languages:** Portuguese (Native), English (Advanced / Fluent)
- **Technical:** AWS serverless architecture, AWS AppSync, AWS Lambda, DynamoDB, DynamoDB Streams, GraphQL subscriptions, Event-driven architecture, IoT telemetry, High-volume data pipelines, Cost-aware cloud design, Observability, Backend architecture, Python, TypeScript, JavaScript, Node.js, Java, PHP, C#, Ruby, C++, SQL, Express, Flask, Django, CherryPy, REST APIs, GraphQL, JWT authentication, SOAP integrations, Serverless architecture, AWS Lambda, AWS AppSync, DynamoDB, DynamoDB Streams, Docker, Terraform, Pulumi, Git, CI/CD, MongoDB, MySQL, SQL Server, React, Next.js, Vue, Quasar, Pinia, Node-RED, RAG, MCP, LLMs, AI agents, Linux, Neovim, tmux

Education

Instituto Federal de Santa Catarina - IFSC

Santa Catarina, Brasil

Análise e Desenvolvimento de Sistemas

Dec 2020 - Present

- Formação superior em desenvolvimento de sistemas, arquitetura de software, bancos de dados e engenharia de software.

Instituto Federal Catarinense - IFC

Santa Catarina, Brasil

Técnico em Informática

Dec 2019 - Dec 2020

- Formação técnica em informática concluída em 2020.